

Module -03-Final Activity – Basic and Advance LAN Security, OSPF, IP Addressing, Network Services, VLAN, EtherChannel, ACL, NAT, NTP and Syslog (08 Hours) 2024.10.05

Devices Name	Interface	Connect with	Interface	User Count	Remark
R-01	Gig 0/0/0	SW-01	Gig 0/1	User 150	
	Gig 0/0/1	SW-02	Gig 0/1	User 30	
	Serial 0/1/0	R-02	Serial 0/1/0	User 02	DTE
R-02	Serial 0/1/0	R-01	Serial 0/1/0		DCE
	Serial 0/1/1	R-03	Serial 0/1/1		DCE
	Serial 0/2/0	R-04	Serial 0/2/0		DCE
	Serial 0/2/1	R-05	Serial 0/2/1		DCE
	Gig 0/0/0	SW-03	Gig 0/1	User 50	
	-	SW-03-1	-		
R-03	Gig 0/0/0	SW-04	Gig 0/1	CCNA	User 50
				CCNP	User 30
				CCIE	User 20
	Gig 0/0/1	SW-05	Gig 0/1	User 100	
	Serial 0/1/1	R-02	Serial 0/1/1	User 02	DTE
R-04	Gig 0/0/0	SW-06	Gig 0/1	User 60	
	Gig 0/0/1	SW-07	Gig 0/1	User 100	
	Serial 0/2/0	R-02	Serial 0/2/0	User 02	DTE
	Serial 0/2/1	ISP	Serial 0/2/1	100.10.20.2/30	DTE
R-05	Gig 0/0/0	SW-08	Gig 0/1	User 120	
	Gig 0/0/1	SW-09	Gig 0/1	User 210	
	Serial 0/2/1	R-02	Serial 0/2/1	User 02	DTE
ISP	Serial 0/2/1	R-04	Serial 0/2/1	100.10.20.1/30	DCE
	Gig 0/0/1	ISP-SW-01	Gig 0/1	100.11.20.1/24	
Amanda-PC	Fa 0	SW-01	Fa 0/1		Amanda Perera
Kasun-PC	Fa 0	SW-01	Fa 0/2		Kasun Jayakody
Dilshan-PC	Fa 0	SW-02	Fa 0/1		Dilshan Herath
Pradeep-PC	Fa 0	SW-02	Fa 0/2		Pradeep Rajapaksha
Lankika-PC	Fa 0	SW-04	Fa 0/1		Lankika Jayathunga
Prabodha-PC	Fa 0	SW-04	Fa 0/2		Prabodha Silva
Gayan-PC	Fa 0	SW-05	Fa 0/1		Gayan Samarasingha
Visura-PC	Fa 0	SW-05	Fa 0/2		Visura Peris
Nayantha-PC	Fa 0	SW-06	Fa 0/1		Nayantha Maleesha
Udara-PC	Fa 0	SW-06	Fa 0/2		Udara Kaushalya
Nadeeka-PC	Fa 0	SW-07	Fa 0/1		Nadeeka Jayasooriya
Binal-PC	Fa 0	SW-07	Fa 0/2		Binal Perera
Hiran-PC	Fa 0	SW-08	Fa 0/1		Hiran Peris
Lahiru-PC	Fa 0	SW-08	Fa 0/2		Lahiru Dias
Web /DHCP	Fa 0	SW-09	Fa 0/1		
DNS / Email	Fa 0	SW-09	Fa 0/2		

01. Using VLSM Method and Finding IP Address. As well as every **GigabytEthernet** Interface and Every **Serial DCE** ports are using **their Lowest IP** address.

i. Basic Configuration (**Every Router**). Set date and time for **every network** device. Set date and time using NTP server.

ii. Set Banner message

```
#####  
This is Secure area. Please enter your Authentication.  
#####
```

iii. Password:

Enable Secret: **Your Name with @2025.lk (Second Letter Capital)**
Console Password: **Your Name with @2025.lk(Third Letter Capital)**

SSH Configuration

Username: Admin
Secret: yourname@Cisco.Lk
RSA Crypto key Module: **1024**
Domain-name: **cisco.lk**

02. Every password **must** be Encrypted.

03. Password minimum length is **11** digits.

04. Blocking Brute-force attack

Failed Attempts - 03

Within - 01minits

Block for - 10minits

05. Create a web site to web server as a Sri Lanka Youth School of Computing. Its domain name www.cisco.lk well as every user can access the web site.

06. Create an email server for all clients and their password is Mail@123. All users have the same email password, and their email address is following like this.

Eg :- Binal@cisco.lk

07. Every remote session and console session's time duration is 10 minutes.

08. DNS server IP address is their Second Lowest address and WEB Server for use Third Lowest address.

10. Save every network device configuration to the TFTP server.

11. Use WEB IP address for TFTP, SYSLOG and NTP.

12. Router-03 Switch 04 has a VLAN Configuration. (Implement VLAN security and allow all Default VLAN and data VLAN through trunk port)

VLAN-02	CCNA	1-5	USERS 50
VLAN-03	CCNP	6-10	USERS 30
VLAN-04	CCIE	11-15	USERS 20

13. Implement OSPF route.

- Process ID 10
- Area 0

14. Shutdown every unwanted port. And implement static MAC Filtering.

15. Established between SW-03 and SW-03-1 to Ether-channel.

SW 03	SW 03-1	Channel No	Channel Method
Fa 0/20, Fa0/21	Fa 0/20, Fa0/21	01	PAgP
Fa 0/22, Fa0/23	Fa 0/22, Fa0/23	02	PAgP

16. Root Bridge must be a not a current one. using priority command

17. Implement Dynamic NAT to Access Internet using this IP Block.

- IP Block: 123.231.1.0/29
- Pool Name: NAT-POOL
- DNS server: 123.231.1.1/29
- WEB Server: 123.231.1.2/29

18. Access Control List.

- SW-04 network users cannot ping server network. But they can access all services.
- SW-01 network cannot access SW-03 and SW-03-1 network.
- Hiran-PC cannot access web service.